

Data Mining Concepts and Techniques (CS24303) — Full Course Master Book

Complete End-to-End B.Tech CSE 5th Semester Study Book & PYQ Bank (46 Topics)

EXECUTIVE MASTER STUDY GUIDE & 46-TOPIC QUESTION BANK

Data Mining Concepts & Techniques (CS24303)

Birla Institute of Technology, Mesra | B.Tech CSE 5th Semester (NEP 2024–25 Scheme)

Complete 46-Topic Course Structure & Syllabus Matrix

Module	Topics	Core Syllabus Scope & Key Algorithms
Module I	14 Topics	KDD Pipeline, 4 Attribute Types, 5-Number Summaries, Proximity Metrics (L1, L2, Minkowski, Cosine, Jaccard)
Module II	6 Topics	Data Cleaning (KNN/MICE/EM), Chi-Square Contingency Test, Normalization, PCA Full Derivation, ChiMerge, 3-4-5 Rule
Module III	9 Topics	Inmon 4 Pillars, OLTP vs OLAP, Star/Snowflake Schemas, 3D Data Cube Lattice, 5 OLAP Operations, BUC, Star-Cubing, AOI
Module IV	7 Topics	Market Basket Analysis, Apriori Downward Closure Proof & 9-Transaction Trace, ECLAT Vertical Format, FP-Growth Recursive Mining, Lift/Kulc
Module V	10 Topics	Multilevel/Multidimensional Rules, 4 Constraint Classes & Proofs, Pattern-Fusion Colossal Mining, Closed vs Maximal Itemsets

Exam Preparation & High-Yield Strategy

This publication-grade master book consolidates all 46 topics with formal KaTeX-rendered mathematical proofs, step-by-step worked traces on standard datasets, full SQL DDL warehouse schemas, and comprehensive model answers to BIT Mesra end-semester examination questions.

Module I: Introduction to Data Mining — Complete 14-Topic Syllabus Tracker

Topic 1: Introduction to Data Mining (Knowledge Discovery Process)

Topic 3: Data Warehouses (SITN Characteristics & OLAP)

Topic 5: Advanced Database Systems & Applications

Topic 7: Classification of Data Mining Systems

Topic 9: Data Concepts (Objects, Attributes & Datasets)

Topic 11: Basic Statistical Descriptions (Mean, Median, Mode, IQR, Box Plots)

Topic 13: Measuring Data Similarity (Cosine Similarity & Dot Products)

Topic 2: Relational Databases (Tables, Tuples & Schemas)

Topic 4: Transactional Databases (Transaction Sets & Market Basket)

Topic 6: Data Mining Functionalities (Characterization vs. Discrimination)

Topic 8: Major Issues in Data Mining (Quality, Scalability, Privacy)

Topic 10: Data Objects & Attribute Types (Nominal, Binary, Ordinal, Numeric)

Topic 12: Data Visualization (Histograms, Scatter Plots, Quantile Plots)

Topic 14: Measuring Data Dissimilarity (Euclidean & Manhattan Distances)

Topic 1 & 2: Introduction to Data Mining & Relational Databases

Data Mining (Knowledge Discovery in Databases — KDD) is the computational process of discovering non-trivial, implicit, previously unknown, and potentially actionable patterns and models from massive datasets.

Database Query (SQL / OLTP)

Retrieves exact known records matching user-specified criteria (e.g., "Find all students with CGPA > 8.0").

Deductive, exact, deterministic SQL responses.

Data Mining (KDD / Discovery)

Discovers hidden, inductive relationships and associations (e.g., "Students who score high in Algorithms also excel in Compiler Design").

Inductive, statistical, pattern-based models with probabilistic confidence.

Topic 3 – 5: Data Warehouses & Advanced Databases [UPLOADED PYQ]

Memory Hook: SITN Characteristics of Data Warehouses (Inmon)

- **S — Subject-Oriented:** Organized around major business subjects (Customers, Products, Sales) rather than transactional operations.
- **I — Integrated:** Data from diverse operational sources is standardized into consistent naming, units, and encoding schemes.
- **T — Time-Variant:** Maintains historical data spanning 5–10 years with explicit timestamp attributes for trend analysis.
- **N — Non-Volatile:** Read-only analytical storage; historical snapshots are loaded and not updated in-place.

Operational Database (OLTP)	Data Warehouse (OLAP) [UPLOADED PYQ — IT335 Mid]
Supports day-to-day operational transactions (e.g., bank ATM withdrawal).	Supports long-term executive decision making, reporting, and predictive analytics.
Stores current, real-time up-to-the-second data.	Stores historical snapshots over long multi-year horizons.
Highly normalized tables (3NF/BCNF) to eliminate update anomalies.	Denormalized dimensional schemas (Star / Snowflake) to optimize complex query joins.

Topic 6: Data Mining Functionalities [UPLOADED PYQ]

Functionality	Core Algorithmic Objective	Illustrative Example
1. Characterization [UPLOADED PYQ]	Summarizes the general properties of a target class into compact descriptions (Attribute-Oriented Induction).	"Summarize the demographic profile of customers who spent > ₹1,00,000 last year."
2. Discrimination [UPLOADED PYQ]	Compares the target class against one or a set of contrasting comparative classes.	"Compare customers who buy gaming laptops vs. customers who buy standard business laptops."
3. Association & Correlation	Finds frequent itemsets and conditional dependency rules ($X \implies Y$).	"Customers purchasing Milk and Bread also buy Butter with 80% confidence."
4. Classification	Supervised learning predicting discrete categorical class labels.	"Classifying credit card transactions as Fraudulent or Legitimate."
5. Prediction (Regression)	Predicts continuous real-valued numeric values.	"Predicting real-estate house prices based on square footage and location."
6. Clustering	Unsupervised grouping of unlabeled data points based on geometric similarity.	"Customer market segmentation into 5 distinct spending behavioral clusters."
7. Outlier Analysis	Detects anomalous objects that deviate significantly from normal baseline distributions.	"Network intrusion detection and credit card fraud surveillance."

Topic 8 – 10: Data Objects & Attribute Types

Attribute Type	Mathematical Characteristics & Valid Operations	Representative Examples
1. Nominal	Categorical labels, names, or codes with no meaningful rank or ordering ($=$, $\neq$).	`Marital_Status` (Single, Married), `Color` (Red, Blue).
2. Binary	Nominal attribute with exactly 2 states (`0` and `1`). Symmetric (Gender) vs. Asymmetric (Medical test positive/negative).	`Smoker` (Yes/No), `COVID_Positive` (0/1).
3. Ordinal	Categorical values with a meaningful ranking / order , but intervals between values cannot be measured ($<$, $>$, $=$, $\neq$).	`Customer_Rating` (Poor < Fair < Good < Excellent), `Academic_Grade` (A, B, C, D, F).
4. Numeric (Interval)	Measured on a scale of equal units; zero is arbitrary (no true zero point; addition and subtraction meaningful, ratios meaningless).	Temperature in $^{\circ}\text{C}$ or $^{\circ}\text{F}$, Calendar Years.
5. Numeric (Ratio)	Inherent absolute zero point (both differences and ratios are mathematically meaningful: $y = 2x$).	`Salary` (₹50,000), `Age` (20 years), `Weight`, `Length`.

Topic 11 & 12: Basic Statistical Descriptions & Visualization

Five-Number Summary & Box-Plot Formulations:

Five-Number Summary = $\langle \text{Minimum}, Q_1 \text{ (25\%)}, \text{Median } (Q_2), Q_3 \text{ (75\%)}, \text{Maximum} \rangle$

Interquartile Range (IQR) = $Q_3 - Q_1$

Outlier Cutoff Boundaries: Lower = $Q_1 - 1.5 \times \text{IQR}$, Upper = $Q_3 + 1.5 \times \text{IQR}$

Topic 13 & 14: Data Similarity & Dissimilarity Measures

1. Cosine Similarity (For Sparse Document/Text Vectors $\mathbf{x}, \mathbf{y}$):

$$\text{sim}(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\|_2 \|\mathbf{y}\|_2} = \frac{\sum_{i=1}^n x_i y_i}{\sqrt{\sum_{i=1}^n x_i^2} \sqrt{\sum_{i=1}^n y_i^2}}$$

2. Minkowski Distance Metric (L_p Norm):

$$d(\mathbf{x}, \mathbf{y}) = \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{\frac{1}{p}}$$

- For $p = 1$: **Manhattan (L_1) Distance**: $d(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^n |x_i - y_i|$ - For $p = 2$: **Euclidean (L_2) Distance**: $d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$

M1 Active Recall & Exam Questions [UPLOADED PYQ]

Q1. [UPLOADED PYQ — IT426 Mid] Differentiate between Data Characterization and Data Discrimination with concrete examples. (8 Marks)

- **Data Characterization:** A summarization of the general characteristics or features of a target class of data. The output can be represented as pie charts, bar charts, multidimensional data cubes, or generalized relations.

Example: Characterizing customers who purchased items worth over ₹1,00,000 at an electronics store to find their average age, income, and city.

- **Data Discrimination:** A comparative analysis of the general features of the target class objects against one or more contrasting comparative classes.

Example: Comparing software engineers who remained at a company for over 5 years against those who left within 1 year to identify key turnover drivers.

Module II: Data Preprocessing — Complete 6-Topic Syllabus Tracker

Topic 15: Data Cleaning (Handling Missing Values & Noisy Data Smoothing)

Topic 17: Data Transformation (Min-Max & Z-Score Normalizations)

Topic 19: Data Discretization (Unsupervised Binning & Supervised Trees)

Topic 16: Data Integration (Schema Matching & Redundancy Correlation)

Topic 18: Data Reduction (Dimensionality, Numerosity & Compression)

Topic 20: Concept Hierarchy Generation for Nominal & Numeric Data

Memory Hook: C-I-T-R-D-H Preprocessing Pipeline

Clean → Integrate → Transform → Reduce → Discretize → Hierarchy

Topic 15: Data Cleaning Techniques

Missing Data Strategy	Operational Mechanism	Tradeoffs & Constraints
1. Ignore Tuple	Discard the entire row containing missing values.	Simple; highly wasteful if tuple contains other valuable attributes.
2. Global Constant	Fill missing values with a label like "Unknown" or $-\infty$.	May mislead data mining classifiers into treating "Unknown" as a true category.
3. Attribute Mean / Median	Replace missing numeric value with mean (symmetric distribution) or median (skewed data).	Preserves overall sample mean; reduces natural variance.
4. Most Probable Value	Estimate using regression, decision tree inference, or Bayesian expectation maximization.	Statistically soundest; computationally intensive.

Topic 17: Data Transformation & Normalization [UPLOADED PYQ]

1. Min-Max Normalization Formula [UPLOADED PYQ]:

$$v' = \frac{v - \min_A}{\max_A - \min_A} (\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A$$

Maps values into designated target range $[\text{new_min}_A, \text{new_max}_A]$ (commonly $[0.0, 1.0]$).

2. Z-Score (Zero-Mean) Normalization Formula:

$$v' = \frac{v - \bar{A}}{\sigma_A}$$

Where $\bar{A}$ is the mean and σ_A is the standard deviation. Essential when actual minimum and maximum are unknown or outliers dominate.

 [UPLOADED PYQ] Solved Problem: Min-Max Normalization of Value 400 in Range [200, 1000] to Target [10, 20]

Given: $v = 400$, $\min = 200$, $\max = 1000$, $\text{new_min} = 10$, $\text{new_max} = 20$.

$$v' = \frac{400 - 200}{1000 - 200} \times (20 - 10) + 10 = \frac{200}{800} \times 10 + 10 = 0.25 \times 10 + 10 = 2.5 + 10 = \mathbf{12.5}$$

Topic 18 & 19: Data Reduction & Discretization

Reduction Approach	Core Mathematical / Statistical Techniques	Key Benefits
1. Dimensionality Reduction	Principal Component Analysis (PCA): Projects d -dimensional space onto k orthogonal eigenvectors ($k \ll d$) maximizing variance. Wavelet transforms.	Removes irrelevant features; mitigates Curse of Dimensionality.
2. Numerosity Reduction	Parametric: Linear regression models ($y = mx + c$), Log-linear models. Non-parametric: Histograms, Clustering, Random Sampling without replacement.	Replaces massive raw dataset with compact mathematical surrogate models.
3. Discretization (Binning)	• Equal-Width (Distance) Partitioning: Divides range into k equal intervals: $W = \frac{\max - \min}{k}$. • Equal-Frequency (Depth) Partitioning: Divides sorted values such that each bin contains N/k samples.	Converts continuous attributes into discrete intervals for Decision Trees and Association Mining.

Topic 20: Concept Hierarchy Generation [UPLOADED PYQ]

[UPLOADED PYQ — IT426 End] Concept Hierarchy Generation for Nominal Data

A **Concept Hierarchy** defines a sequence of mappings from a set of low-level concepts to higher-level, more general concepts:

Street $\longrightarrow$ City $\longrightarrow$ State $\longrightarrow$ Country

Two Primary Methods of Generation:

1. **Specification of Partial Ordering by Users / Experts at Schema Level:** Explicitly defined by database administrators (e.g., `street < city < state < country`).
2. **Data-Driven Automatic Generation Based on Attribute Value Counts (Distinct Cardinality):** The attribute with the largest number of distinct values is placed at the lowest level of the hierarchy, while the attribute with the fewest distinct values is placed at the top (e.g., `Country` (10 values) > `State` (50 values) > `City` (500 values) > `Street` (50,000 values)).

M2 Active Recall & Exam Questions [UPLOADED PYQ]

Q1. [UPLOADED PYQ] Explain Equal-Width vs. Equal-Frequency Binning for data smoothing on sorted data: `4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34`. (8 Marks)

Dataset size $N = 12$, target bins $k = 3$:

- **Equal-Frequency (Depth) Partitioning (4 elements per bin):**

- Bin 1: `4, 8, 9, 15` → Smoothed by bin mean: `9, 9, 9, 9`
- Bin 2: `21, 21, 24, 25` → Smoothed by bin mean: `23, 23, 23, 23`
- Bin 3: `26, 28, 29, 34` → Smoothed by bin mean: `29, 29, 29, 29`

- **Equal-Width Partitioning:** Width $W = \frac{34-4}{3} = 10$.

- Bin 1 $[4, 14)$: `4, 8, 9`
- Bin 2 $[14, 24)$: `15, 21, 21`
- Bin 3 $[24, 34]$: `24, 25, 26, 28, 29, 34`

Module III: Data Warehouse & OLAP Technology — Complete 9–Topic Syllabus Tracker

Topic 21: Basic Concepts of Data Warehouse & ETL Pipelines

Topic 23: Data Cube Multidimensional Views (Base to Apex Cuboids)

Topic 25: Data Warehouse Design & 4 Architecture Views [UPLOADED PYQ]

Topic 27: Attribute–Oriented Induction (AOI Generalization Algorithm)

Topic 29: Preliminary Warehouse Concepts (Fact vs. Dimension Tables)

Topic 22: Data Warehouse Modeling (Star, Snowflake, Fact Constellation)

Topic 24: OLAP Operations (Roll–Up, Drill–Down, Slice, Dice, Pivot)

Topic 26: Data Warehouse Implementation (Materialized Views & Indexing)

Topic 28: Data Cube Computation & Materialization Choices [UPLOADED PYQ]

Topic 21 & 22: Data Warehouse Modeling Schemas [UPLOADED PYQ]

Dimensional Schema	Structural Layout & Normalization	Key Advantages & Tradeoffs
1. Star Schema	Single central Fact Table connected radially to completely denormalized Dimension Tables (1–join query topology).	Simplest structure; blazing fast OLAP query response times. Contains data redundancy in dimension tables.
2. Snowflake Schema [UPLOADED PYQ]	Dimension tables are normalized into multiple hierarchical sub-tables (e.g., `Product` splits into `Product_Category` and `Supplier`).	Zero data redundancy; saves storage space. Slower query execution due to expensive multi-table relational joins.
3. Fact Constellation (Galaxy)	Multiple Fact Tables (e.g., `Sales_Fact` and `Shipping_Fact`) sharing common conformed Dimension Tables (`Time`, `Item`).	Models sophisticated enterprise-wide business operations across multiple departments.

Topic 23 & 24: Data Cube & OLAP Operations [UPLOADED PYQ]

The 5 Canonical OLAP Operations

1. **Roll–Up (Drill–Up):** Aggregates data by climbing up a concept hierarchy or reducing dimensions (e.g., from `Day` → `Month` → `Quarter` → `Year`).
2. **Drill–Down:** Navigates from summarized high-level data to detailed granular data (e.g., from `Country` → `State` → `City`).
3. **Slice:** Performs a selection on a single dimension, producing a 2D sub-cube (e.g., `Time` = "Q1 2026").
4. **Dice:** Defines a sub-cube by performing a selection on two or more dimensions (e.g., `Location` = {"Ranchi", "Patna"} AND `Item` = {"Laptop", "Phone"}).
5. **Pivot (Rotate):** Rotates the data axes in visualization to provide alternative multidimensional viewing angles.

Feature	OLTP (Online Transaction Processing)	OLAP (Online Analytical Processing) [UPLOADED PYQ]
Primary Focus	Day-to-day transaction processing and CRUD operations.	Complex multidimensional decision support and data mining.
Data Content	Current, real-time snapshot; highly detailed.	Historical, consolidated, multi-year summarized aggregates.
Database Design	Highly normalized entity-relationship (ER) models (3NF).	Denormalized multidimensional Star / Snowflake schemas.
Access Pattern	High frequency, short, deterministic atomic transactions.	Low frequency, complex analytical aggregation queries.

Topic 25 – 28: Warehouse Design Views & Cube Materialization [UPLOADED PYQ]

[UPLOADED PYQ – IT426 Mid] The 4 Essential Design Views for a Data Warehouse

1. **Top-Down View:** Defines the overall information requirements of decision makers and enterprise business goals.
2. **Data Source View:** Models the operational schemas, data types, and access protocols of legacy source systems.
3. **Data Warehouse View:** Defines the dimensional Star/Snowflake schemas, fact measures, dimensions, and aggregation granularities.
4. **Business Query View:** Analyzes user query patterns, analytical workloads, and reporting requirements to guide index design and cube materialization.

Data Cube Materialization Choices (2^n Cuboids) [UPLOADED PYQ]:

- **1. No Materialization:** Compute cuboids on-the-fly when requested. Zero disk overhead; extremely slow query latency.
- **2. Full Materialization:** Precompute all 2^n possible cuboids in the lattice. Fast query response; massive storage overhead.
- **3. Partial Materialization (Optimal Industry Standard):** Precomputes a carefully selected subset of high-frequency cuboids based on query workload and storage constraints (Greedy algorithm / PBS).

M3 Active Recall & Exam Questions [UPLOADED PYQ]

Q1. [UPLOADED PYQ] Compare Star Schema and Snowflake Schema across 4 architectural parameters. (8 Marks)

1. **Normalization Level:** Star schema uses denormalized dimension tables; Snowflake schema normalizes dimension tables into 3NF hierarchies.
2. **Query Performance:** Star schema provides faster query execution because queries require only simple single-level joins with the central fact table; Snowflake requires multiple expensive joins across normalized dimension sub-tables.
3. **Storage Overhead:** Star schema contains data redundancy (e.g., repeated city/state names); Snowflake eliminates redundancy, saving disk space.
4. **Maintenance Complexity:** Star schema is easier to understand and maintain for business analysts; Snowflake has complex structural dependencies.

Module IV: Frequent Pattern & Association Rule Mining — Complete 7-Topic Tracker

Topic 30: Basic Concepts of Frequent Pattern Mining

Topic 32: Correlation Measures (Lift & Chi-Square Analysis) [UPLOADED PYQ]

Topic 34: The Apriori Algorithm (Join, Prune & Improvements) [UPLOADED PYQ]

Topic 36: Interesting Pattern Evaluation Measures (S-C-L Triad)

Topic 31: Association Rules (Support & Confidence Metrics)

Topic 33: Frequent Itemset Mining Methods Overview

Topic 35: The FP-Growth Algorithm (FP-Tree Construction & Mining) [UPLOADED PYQ]

Topic 30 & 31: Association Rule Mining Formulations

1. Support and Confidence Formulations for Rule $A \implies B$:

$$\text{Support}(A \implies B) = P(A \cup B) = \frac{\text{Count}(A \cup B)}{|D|}$$

$$\text{Confidence}(A \implies B) = P(B | A) = \frac{\text{Support}(A \cup B)}{\text{Support}(A)} = \frac{\text{Count}(A \cup B)}{\text{Count}(A)}$$

A rule is **Strong** if $\text{Support} \geq \text{min_sup}$ and $\text{Confidence} \geq \text{min_conf}$.

Topic 32: Correlation Measures & The Lift Metric [UPLOADED PYQ]

Lift Correlation Formula [UPLOADED PYQ – IT426 Mid]:

$$\text{Lift}(A \implies B) = \frac{P(A \cup B)}{P(A) \cdot P(B)} = \frac{\text{Confidence}(A \implies B)}{\text{Support}(B)}$$

- $\text{Lift} > 1$: Positive correlation (A and B occur together more frequently than expected by chance).
- $\text{Lift} = 1$: Statistical independence (no correlation between A and B).
- $\text{Lift} < 1$: Negative correlation (the presence of A discourages the occurrence of B).

Topic 34: The Apriori Algorithm (Join & Prune Steps) [UPLOADED PYQ]

The Apriori Property & Pruning Theorem (Agrawal & Srikant, 1994)

Apriori Invariant: All non-empty subsets of a frequent itemset must also be frequent.

Anti-Monotonicity Pruning Rule: If an itemset is infrequent, all of its supersets are guaranteed to be infrequent and can be immediately pruned without scanning the database!

1. **Join Step:** C_k candidate itemsets are generated by joining L_{k-1} with itself ($L_{k-1} \bowtie L_{k-1}$).
2. **Prune Step:** Any candidate $c \in C_k$ is deleted if any of its $(k-1)$ -subsets is not present in L_{k-1} .

Topic 35: The FP-Growth Algorithm [UPLOADED PYQ]

FP-Growth (Frequent Pattern Growth) overcomes the major computational bottlenecks of Apriori (repeated database scans and massive candidate generation) by constructing a compact prefix-tree structure called the **FP-Tree**.

 **[UPLOADED PYQ — IT335 End] Full Solved Numerical: FP-Growth on Transaction Database ($\text{min_sup} = 60\% = 3$ out of 5)**

Given 5 Transactions:

TID	Items Purchased	Ordered Frequent Items (Support ≥ 3)
10	Beer, Nuts, Diapers	$\langle \text{Nuts} : 4, \text{Beer} : 4, \text{Diapers} : 4 \rangle$
20	Beer, Coffee, Diapers, Nuts	$\langle \text{Nuts} : 4, \text{Beer} : 4, \text{Diapers} : 4 \rangle$ (Coffee dropped)
30	Beer, Diapers, Eggs	$\langle \text{Beer} : 4, \text{Diapers} : 4, \text{Eggs} : 3 \rangle$
40	Beer, Nuts, Eggs, Milk	$\langle \text{Nuts} : 4, \text{Beer} : 4, \text{Eggs} : 3 \rangle$ (Milk dropped)
50	Nuts, Coffee, Diapers, Eggs, Milk	$\langle \text{Nuts} : 4, \text{Diapers} : 4, \text{Eggs} : 3 \rangle$

Step 1: Frequent 1-Itemsets: $\{\text{Nuts} : 4, \text{Beer} : 4, \text{Diapers} : 4, \text{Eggs} : 3\}$. Frequency Order: $\text{Nuts} \rightarrow \text{Beer} \rightarrow \text{Diapers} \rightarrow \text{Eggs}$.

Step 2: Mining Conditional Pattern Bases from Bottom of Header Table:

- **Item 'Eggs':** Conditional Pattern Base: $\{\langle \text{Beer}, \text{Diapers} : 1 \rangle, \langle \text{Nuts}, \text{Beer} : 1 \rangle, \langle \text{Nuts}, \text{Diapers} : 1 \rangle\}$. Conditional FP-Tree: $\emptyset$ (No item has support ≥ 3). Frequent Itemset: $\{\text{Eggs} : 3\}$.
- **Item 'Diapers':** Conditional Pattern Base: $\{\langle \text{Nuts}, \text{Beer} : 2 \rangle, \langle \text{Beer} : 1 \rangle, \langle \text{Nuts} : 1 \rangle\}$. Conditional FP-Tree: $\langle \text{Nuts} : 3, \text{Beer} : 3 \rangle$. Frequent Itemsets: $\{\text{Diapers}, \text{Nuts} : 3, \text{Diapers}, \text{Beer} : 3\}$.
- **Item 'Beer':** Conditional Pattern Base: $\{\langle \text{Nuts} : 3 \rangle\}$. Conditional FP-Tree: $\langle \text{Nuts} : 3 \rangle$. Frequent Itemset: $\{\text{Beer}, \text{Nuts} : 3\}$.

M4 Active Recall & Exam Questions [UPLOADED PYQ]

Q1. [UPLOADED PYQ — IT426 Mid] Compare the significance of Support and Confidence in association analysis. Why is Lift necessary? (8 Marks)

- **Support:** Measures how frequently an itemset appears across the entire dataset ($P(A \cup B)$). It filters out statistically insignificant random noise.
- **Confidence:** Measures the conditional reliability of the inference rule ($P(B | A)$). High confidence means B is likely when A occurs.
- **Why Lift is Necessary:** Support and Confidence alone can be deceptive. For example, if 90% of all customers buy Tea ($P(\text{Tea}) = 0.9$) and rule 'Coffee $\Rightarrow$ Tea' has 80% confidence, Coffee actually *decreases* the likelihood of buying Tea! Lift computes $\frac{0.80}{0.90} = 0.89 < 1$, exposing the negative correlation that Confidence hid.

Module V: Advanced Pattern Mining, Classification & Clustering – Complete Tracker

Topic 37: Pattern Mining Road Map (Itemsets to Multidimensional)	Topic 38: Multilevel Pattern Mining (Concept Hierarchy Abstractions)
Topic 39: Multidimensional Pattern Mining (Multiple Predicates)	Topic 40: Constraint-Based Pattern Mining (Pushing Constraints)
Topic 41: Mining High-Dimensional Data & Colossal Patterns	Topic 42: Mining Compressed Patterns (Closed vs. Maximal Itemsets)
Topic 43: Mining Approximate Patterns & Error Tolerances	Topic 44 & 45: Pattern Exploration & Real-World Domain Applications
PYQ Block A: Classification (Decision Trees, Naïve Bayes, Backprop & Metrics)	PYQ Block B: Clustering (K-Means, PAM Medoids, DBSCAN & BIRCH CF-Trees)

Topic 37 – 40: Advanced Pattern Mining Methodologies

Advanced Mining Paradigm	Algorithmic Mechanism & Description	Key Benefit / Example
1. Multilevel Association	Mines rules across different concept hierarchy levels (e.g., `Milk => Bread` vs. `Dairy => Baked Goods`).	Uncovers granular patterns at lower levels and generalized trends at higher levels.
2. Multidimensional Association	Involves multiple predicates or dimensions (e.g., `age(X, "20..29") ^ income(X, "50K..60K") => buys(X, "Laptop")`).	Discovers rich customer demographic profiles beyond simple single-item co-occurrences.
3. Constraint-Based Mining	Pushes user-defined constraints (Antimonotonic, Monotonic, Succinct) directly into the mining algorithm.	Massively prunes the candidate search space; produces only actionable, relevant rules.
4. Closed vs. Maximal Patterns	• Closed Itemset: An itemset X is closed if no proper superset has the same support. • Maximal Itemset: An itemset X is maximal if no proper superset is frequent.	Compresses millions of frequent itemsets into compact loss-less or lossy representations.

★ PYQ–CRITICAL BLOCK A: Classification Algorithms [UPLOADED PYQ]

1. Naïve Bayesian Classification [UPLOADED PYQ – IT426 End]:

$$P(C_i \mid \mathbf{X}) = \frac{P(\mathbf{X} \mid C_i)P(C_i)}{P(\mathbf{X})} = \frac{P(C_i) \prod_{k=1}^n P(x_k \mid C_i)}{P(\mathbf{X})}$$

Class-Conditional Independence Assumption: Assumes that the effect of an attribute value on a given class is completely independent of the values of other attributes.

2. Backpropagation Neural Network Learning [UPLOADED PYQ – IT335 End]:

$$w_{ij} \leftarrow w_{ij} + \Delta w_{ij} = w_{ij} + (l)\text{Err}_j O_i$$

- For output layer neuron k : $\text{Err}_k = O_k(1 - O_k)(T_k - O_k)$ - For hidden layer neuron j : $\text{Err}_j = O_j(1 - O_j) \sum_k \text{Err}_k w_{jk}$

3. Classification Evaluation Metrics [UPLOADED PYQ]:

Metric	Formula	Practical Interpretation
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	Overall proportion of correct predictions.
Precision	$\frac{TP}{TP+FP}$	Out of all predicted positives, how many are truly positive?
Recall (Sensitivity)	$\frac{TP}{TP+FN}$	Out of all actual positive cases, how many were discovered?
Error Rate	$\frac{FP+FN}{TP+TN+FP+FN} = 1 - \text{Accuracy}$	Proportion of misclassified instances.

★ PYQ–CRITICAL BLOCK B: Clustering Algorithms [UPLOADED PYQ]

Algorithm [UPLOADED PYQ]	Clustering Paradigm	Algorithmic Principle	Key Advantages & Tradeoffs
1. K-Means [UPLOADED PYQ]	Partitioning (Centroid-based)	Iteratively assigns objects to nearest mean centroid μ_j and recalculates centroids to minimize $SSE = \sum \sum \ x - \mu_j\ ^2$.	Fast $O(tkn)$; sensitive to noise and outliers; requires k upfront.
2. PAM (K-Medoids) [UPLOADED PYQ]	Partitioning (Medoid-based)	Uses actual representative data points (medoids) instead of mean averages. Iteratively evaluates quality swaps.	Highly robust to noise and outliers; computationally expensive $O(k(n - k)^2)$.
3. DBSCAN [UPLOADED PYQ]	Density-Based	Grows clusters from high-density core points ($\geq \text{MinPts}$ within ϵ -radius). Marks sparse points as noise.	Discovers arbitrary-shaped clusters; immune to noise; doesn't require k upfront.
4. BIRCH [UPLOADED PYQ]	Hierarchical (Feature Tree)	Builds an in-memory CF-Tree (Clustering Feature Tree) where each node contains $CF = (N, \text{LS}, SS)$.	Linearly scalable $O(n)$ for massive datasets with single scan of disk.

M5 Active Recall & Exam Questions [UPLOADED PYQ]

Q1. [UPLOADED PYQ — IT335 End] Explain DBSCAN clustering algorithm with core points, border points, and noise points. (8 Marks)

DBSCAN relies on two parameters: radius ϵ and density threshold MinPts :

1. **Core Point:** A point p is a core point if $|N_\epsilon(p)| \geq \text{MinPts}$ (at least MinPts points lie within distance ϵ).
2. **Border Point:** A point q is not a core point, but falls within the ϵ -neighborhood of some core point p .
3. **Noise Point:** Any point that is neither a core point nor a border point.

Clustering: Connects all density-reachable core points into clusters and assigns border points to their adjacent core clusters.